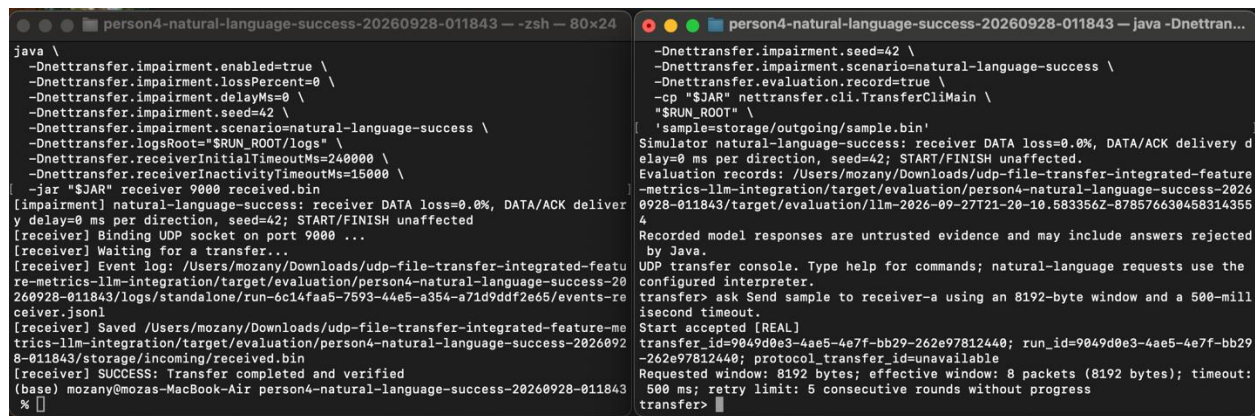


LLM-Assisted Transfer and Demonstration Evidence

This document provides supporting evidence for the LLM-assisted and demonstration requirements of the system. It documents a successful transfer initiated through natural language, evidence-grounded LLM analysis using recorded metrics, deterministic rejection of an invalid command, and a successful transfer under a controlled 2% packet-loss impairment.



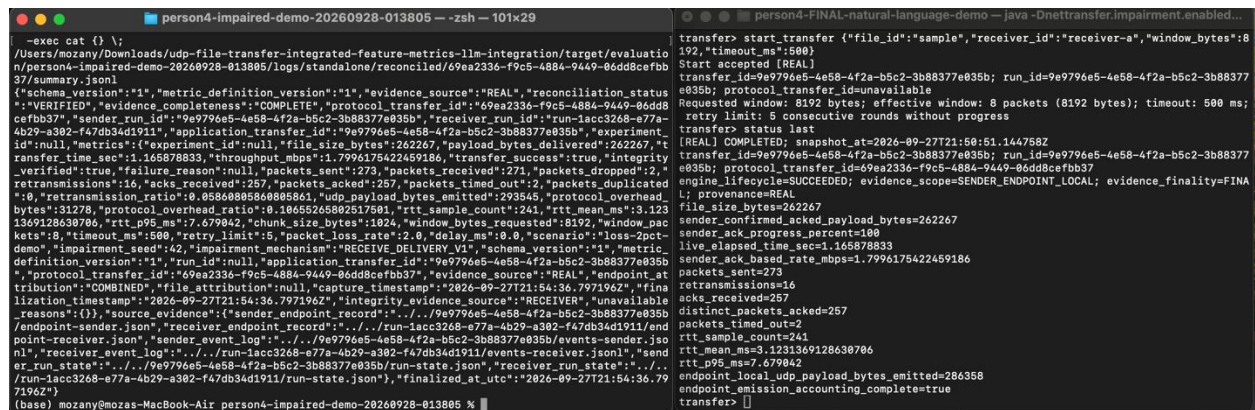
```
java \
-Dnettransfer.impairment.enabled=true \
-Dnettransfer.impairment.lossPercent=0 \
-Dnettransfer.impairment.delayMs=0 \
-Dnettransfer.impairment.seed=42 \
-Dnettransfer.impairment.scenario=natural-language-success \
-Dnettransfer.logsRoot="$RUN_ROOT/logs" \
-Dnettransfer.receiverInitialTimeoutMs=240000 \
-Dnettransfer.receiverInactivityTimeoutMs=15000 \
-jar "$JAR" receiver 9000 received.bin

[impairment] natural-language-success: receiver DATA loss=0.0%, DATA/ACK delivery delay=0 ms per direction, seed=42; START/FINISH unaffected
[receiver] Binding UDP socket on port 9000 ...
[receiver] Waiting for a transfer...
[receiver] Event log: /Users/mozany/Downloads/udp-file-transfer-integrated-feature-metrics-llm-integration/target/evaluation/person4-natural-language-success-20260928-011843/logs/standalone/run-6c14faa5-7593-44e5-a354-a71d9ddf2e65/events-receiver.jsonl
[receiver] Saved /Users/mozany/Downloads/udp-file-transfer-integrated-feature-metrics-llm-integration/target/evaluation/person4-natural-language-success-20260928-011843/storage/incoming/received.bin
[receiver] SUCCESS: Transfer completed and verified
(base) mozany@mozas-MacBook-Air person4-natural-language-success-20260928-011843 %

-Dnettransfer.impairment.seed=42 \
-Dnettransfer.impairment.scenario=natural-language-success \
-Dnettransfer.evaluation.record=true \
-cp "$JAR" nettransfer.cli.TransferCliMain \
"$RUN_ROOT" \
'sample=storage/outgoing/sample.bin'
Simulator natural-language-success: receiver DATA loss=0.0%, DATA/ACK delivery delay=0 ms per direction, seed=42; START/FINISH unaffected.
Evaluation records: /Users/mozany/Downloads/udp-file-transfer-integrated-feature-metrics-llm-integration/target/evaluation/person4-natural-language-success-20260928-011843/target/evaluation/llm-2026-09-27T21-20-10.583356Z-8785766304583143554
Recorded model responses are untrusted evidence and may include answers rejected by Java.
UDP transfer console. Type help for commands; natural-language requests use the configured interpreter.
transfer> ask Send sample to receiver-a using an 8192-byte window and a 500-millisecond timeout.
Start accepted [REAL]
Transfer id=9049d0e3-4ae5-4e7f-bb29-262e97812440; run_id=9049d0e3-4ae5-4e7f-bb29-262e97812440; protocol_transfer_id=unavailable
Requested window: 8192 bytes; effective window: 8 packets (8192 bytes); timeout: 500 ms; retry limit: 5 consecutive rounds without progress
transfer>
```

Figure 1: Natural-Language Demo Environment Setup

The final natural-language demonstration was performed using the integrated receiver and command console. The receiver was started with the configured evaluation settings and waited for the incoming transfer, while the command console was launched separately to allow the transfer to be initiated through the LLM-assisted interface.



```
[-exec cat {} \;
/Users/mozany/Downloads/udp-file-transfer-integrated-feature-metrics-llm-integration/target/evaluation/person4-impaired-demo-20260928-013805/logs/standalone/reconciled/69ea2336-f9c5-4884-9449-86dd8cefb37/summary.jsonl
{"schema_version":"1","metric_definition_version":"1","evidence_source":"REAL","reconciliation_status":"VERIFIED","evidence_completeness":"COMPLETE","protocol_transfer_id":"69ea2336-f9c5-4884-9449-86dd8cefb37","sender_run_id":"9e9796e5-4e58-4f2a-b5c2-3b88377e835b","receiver_run_id":"run-lacc3268-e77a-4b29-a302-f47db34d1911","application_transfer_id":"9e9796e5-4e58-4f2a-b5c2-3b88377e835b","experiment_id":null,"metrics":{"experiment_id":null,"file_size_bytes":262267,"payload_bytes_delivered":262267,"transfer_time_sec":1.165878833,"throughput_mbps":1.7996175422459186,"transfer_success":true,"integrity_verified":true,"failure_reason":null,"packets_sent":273,"packets_received":271,"packets_dropped":2,"retransmissions":16,"acks_received":257,"packets_acked":257,"packets_timed_out":2,"packets_duplicated":0,"retransmission_ratio":0.0586888580885861,"udp_payload_bytes_emitted":293545,"protocol_overhead_bytes":31278,"protocol_overhead_ratio":0.18659265802517581,"rtt_sample_count":241,"rtt_mean_ms":3.1231369128630706,"rtt_p95_ms":7.679042,"chunk_size_bytes":1824,"window_bytes_requested":8192,"window_packets":8,"timeout_ms":500,"retry_limit":5,"packet_loss_rate":2.0,"delay_ms":0.0,"scenario":"loss-2pct-demo","impairment_seed":42,"impairment_mechanism":"RECEIVE_DELIVERY_V1","schema_version":"1","metric_definition_version":"1","run_id":null,"application_transfer_id":"9e9796e5-4e58-4f2a-b5c2-3b88377e835b","protocol_transfer_id":"69ea2336-f9c5-4884-9449-86dd8cefb37","evidence_source":"REAL","reconciliation_status":"COMBINED","file_attribution":null,"capture_timestamp":"2026-09-27T21:54:36.797196Z","finalization_timestamp":"2026-09-27T21:54:36.797196Z","integrity_evidence_source":"RECEIVER","unavailable_reasons":{},"source_evidence":{"sender_endpoint_record":".../9e9796e5-4e58-4f2a-b5c2-3b88377e835b/endpoint-sender.json","receiver_endpoint_record":".../run-lacc3268-e77a-4b29-a302-f47db34d1911/endpoint-receiver.json","sender_event_log":".../9e9796e5-4e58-4f2a-b5c2-3b88377e835b/events-sender.json","receiver_event_log":".../run-lacc3268-e77a-4b29-a302-f47db34d1911/events-receiver.json"},"sender_run_state":".../9e9796e5-4e58-4f2a-b5c2-3b88377e835b/run-state.json","receiver_run_state":".../run-lacc3268-e77a-4b29-a302-f47db34d1911/run-state.json"},"finalized_at_utc":"2026-09-27T21:54:36.797196Z"}
(base) mozany@mozas-MacBook-Air person4-impaired-demo-20260928-013805 %

transfer> start_transfer {"file_id":"sample","receiver_id":"receiver-a","window_bytes":8192,"timeout_ms":500}
Start accepted [REAL]
Transfer id=9e9796e5-4e58-4f2a-b5c2-3b88377e835b; run_id=9e9796e5-4e58-4f2a-b5c2-3b88377e835b; protocol_transfer_id=unavailable
Requested window: 8192 bytes; effective window: 8 packets (8192 bytes); timeout: 500 ms; retry limit: 5 consecutive rounds without progress
transfer> status last
[REAL] COMPLETED; snapshot_at=2026-09-27T21:58:51.144758Z
Transfer id=9e9796e5-4e58-4f2a-b5c2-3b88377e835b; run_id=9e9796e5-4e58-4f2a-b5c2-3b88377e835b; protocol_transfer_id=69ea2336-f9c5-4884-9449-86dd8cefb37
engine_lifecycle=SUCCEEDED; evidence_scope=SENDER_ENDPOINT_LOCAL; evidence_finality=FINAL; provenance=REAL
file_size_bytes=262267
sender_confirmed_acked_payload_bytes=262267
sender_ack_progress_percent=100
live_elapsed_time_sec=1.165878833
sender_ack_based_rate_mbps=1.7996175422459186
packets_sent=273
retransmissions=16
acks_received=257
distinct_packets_acked=257
packets_timed_out=2
rtt_sample_count=241
rtt_mean_ms=3.1231369128630706
rtt_p95_ms=7.679042
endpoint_local_udp_payload_bytes_emitted=286358
endpoint_emission_accounting_complete=true
transfer>
```

Figure 2: Natural-Language Transfer Initiation

The file transfer was initiated using the natural-language request, “Send sample to receiver-a using an 8192-byte window and a 500-millisecond timeout.” The LLM translated the request into the structured start_transfer command, and the Java-controlled interface accepted the request as a REAL transfer. The requested 8192-byte window corresponded to an effective window of eight packets.

```

3885/storage/incoming/received.bin
(base) mozany@mozas-MacBook-Air person4-impaired-demo-20260928-013805 % SENDER_DIR=$(dirname "$(find
$SRUN_ROOT/logs/standalone" -name endpoint-sender.json -type f | head -1)")
RECEIVER_DIR=$(dirname "$(find "$SRUN_ROOT/logs/standalone" -name endpoint-receiver.json -type f | hea
d -1)")

echo "SENDER=$SENDER_DIR"
echo "RECEIVER=$RECEIVER_DIR"
SENDER=/Users/mozany/Downloads/udp-file-transfer-integrated-feature-metrics-llm-integration/target/ev
aluation/person4-impaired-demo-20260928-013805/logs/standalone/9e9796e5-4e58-4f2a-b5c2-3b88377e035b
RECEIVER=/Users/mozany/Downloads/udp-file-transfer-integrated-feature-metrics-llm-integration/target/
evaluation/person4-impaired-demo-20260928-013805/logs/standalone/run-1acc3268-e77a-4b29-a302-f47db34d
1911
(base) mozany@mozas-MacBook-Air person4-impaired-demo-20260928-013805 % java -jar "$JAR" reconcile "$S
ENDER_DIR" "$RECEIVER_DIR"
WARNING: Final field chunkSizeBytes in class nettransfer.metrics.TransferConfiguration has been mutat
ed reflectively by class com.google.gson.internal.bind.ReflectiveTypeAdapterFactory$2 in unnamed modu
le @7b7cd12c (file:/Users/mozany/Downloads/udp-file-transfer-integrated-feature-metrics-llm-integrati
on/target/udp-file-transfer.jar)
WARNING: Use --enable-final-field-mutation=ALL-UNNAMED to avoid a warning
WARNING: Mutating final fields will be blocked in a future release unless final field mutation is ena
bled
[metrics] Summary: /Users/mozany/Downloads/udp-file-transfer-integrated-feature-metrics-llm-integrati
on/target/evaluation/person4-impaired-demo-20260928-013805/logs/standalone/reconciled/69ea2336-f9c5-4
884-9449-06dd8cefb37/summary.json1
[metrics] Manifest: /Users/mozany/Downloads/udp-file-transfer-integrated-feature-metrics-llm-integrati
on/target/evaluation/person4-impaired-demo-20260928-013805/logs/standalone/reconciled/69ea2336-f9c5-
4884-9449-06dd8cefb37/manifest.json
(base) mozany@mozas-MacBook-Air person4-impaired-demo-20260928-013805 %
transfer> start_transfer {"file_id":"sample","receiver_id":"receiver-a","window_bytes":8
192,"timeout_ms":500}
Start accepted [REAL]
transfer_id=9e9796e5-4e58-4f2a-b5c2-3b88377e035b; run_id=9e9796e5-4e58-4f2a-b5c2-3b88377
e035b; protocol_transfer_id=unavailable
Requested window: 8192 bytes; effective window: 8 packets (8192 bytes); timeout: 500 ms;
retry limit: 5 consecutive rounds without progress
transfer> status last
[REAL] COMPLETED; snapshot_at=2026-09-27T21:50:51.144758Z
transfer_id=9e9796e5-4e58-4f2a-b5c2-3b88377e035b; run_id=9e9796e5-4e58-4f2a-b5c2-3b88377
e035b; protocol_transfer_id=unavailable
engine_lifecycle=SUCCEEDED; evidence_scope=SENDER_ENDPOINT_LOCAL; evidence_finality=FINA
L; provenance=REAL
file_size_bytes=262267
sender_confirmed_acked_payload_bytes=262267
sender_ack_progress_percent=100
live_elapsed_time_sec=1.165878833
sender_ack_based_rate_mbps=1.7996175422459186
packets_sent=273
retransmissions=16
acks_received=257
distinct_packets_acked=257
packets_timed_out=2
rtt_sample_count=241
rtt_mean_ms=3.1231369128630706
rtt_p95_ms=7.679842
endpoint_local_udp_payload_bytes_emitted=286358
endpoint_emission_accounting_complete=true
transfer>

```

Figure 3: Successful Natural-Language File Transfer

The receiver successfully completed and verified the file transfer initiated through the natural-language interface. The sender also reported the transfer as completed successfully, demonstrating that the LLM-assisted command path could initiate the actual UDP file-transfer implementation rather than a simulated transfer.

```

[receiver] Waiting for a transfer...
[receiver] Event log: /Users/mozany/Downloads/udp-file-transfer-integrated-feature-metrics-llm-integr
ation/target/evaluation/person4-impaired-demo-20260928-013805/logs/standalone/run-1acc3268-e77a-4b29-
a302-f47db34d1911/events-receiver.json1
[receiver] Saved /Users/mozany/Downloads/udp-file-transfer-integrated-feature-metrics-llm-integrati
on/target/evaluation/person4-impaired-demo-20260928-013805/storage/incoming/received.bin
[receiver] SUCCESS: Transfer completed and verified
(base) mozany@mozas-MacBook-Air person4-impaired-demo-20260928-013805 % shasum -a 256 \
"$SRUN_ROOT/storage/outgoing/sample.bin" \
"$SRUN_ROOT/storage/incoming/received.bin"
873b229250a56ebc54af3b98f63d48608d1207d3262a89f08ce52ce13245f4 /Users/mozany/Downloads/udp-file-tr
ansfer-integrated-feature-metrics-llm-integration/target/evaluation/person4-impaired-demo-20260928-01
3805/storage/outgoing/sample.bin
873b229250a56ebc54af3b98f63d48608d1207d3262a89f08ce52ce13245f4 /Users/mozany/Downloads/udp-file-tr
ansfer-integrated-feature-metrics-llm-integration/target/evaluation/person4-impaired-demo-20260928-01
3805/storage/incoming/received.bin
(base) mozany@mozas-MacBook-Air person4-impaired-demo-20260928-013805 % SENDER_DIR=$(dirname "$(find
$SRUN_ROOT/logs/standalone" -name endpoint-sender.json -type f | head -1)")
RECEIVER_DIR=$(dirname "$(find "$SRUN_ROOT/logs/standalone" -name endpoint-receiver.json -type f | hea
d -1)")

echo "SENDER=$SENDER_DIR"
echo "RECEIVER=$RECEIVER_DIR"
SENDER=/Users/mozany/Downloads/udp-file-transfer-integrated-feature-metrics-llm-integration/target/ev
aluation/person4-impaired-demo-20260928-013805/logs/standalone/9e9796e5-4e58-4f2a-b5c2-3b88377e035b
RECEIVER=/Users/mozany/Downloads/udp-file-transfer-integrated-feature-metrics-llm-integration/target/
evaluation/person4-impaired-demo-20260928-013805/logs/standalone/run-1acc3268-e77a-4b29-a302-f47db34d
1911
(base) mozany@mozas-MacBook-Air person4-impaired-demo-20260928-013805 %
transfer> start_transfer {"file_id":"sample","receiver_id":"receiver-a","window_bytes":8
192,"timeout_ms":500}
Start accepted [REAL]
transfer_id=9e9796e5-4e58-4f2a-b5c2-3b88377e035b; run_id=9e9796e5-4e58-4f2a-b5c2-3b88377
e035b; protocol_transfer_id=unavailable
Requested window: 8192 bytes; effective window: 8 packets (8192 bytes); timeout: 500 ms;
retry limit: 5 consecutive rounds without progress
transfer> status last
[REAL] COMPLETED; snapshot_at=2026-09-27T21:50:51.144758Z
transfer_id=9e9796e5-4e58-4f2a-b5c2-3b88377e035b; run_id=9e9796e5-4e58-4f2a-b5c2-3b88377
e035b; protocol_transfer_id=unavailable
engine_lifecycle=SUCCEEDED; evidence_scope=SENDER_ENDPOINT_LOCAL; evidence_finality=FINA
L; provenance=REAL
file_size_bytes=262267
sender_confirmed_acked_payload_bytes=262267
sender_ack_progress_percent=100
live_elapsed_time_sec=1.165878833
sender_ack_based_rate_mbps=1.7996175422459186
packets_sent=273
retransmissions=16
acks_received=257
distinct_packets_acked=257
packets_timed_out=2
rtt_sample_count=241
rtt_mean_ms=3.1231369128630706
rtt_p95_ms=7.679842
endpoint_local_udp_payload_bytes_emitted=286358
endpoint_emission_accounting_complete=true
transfer>

```

Figure 4: Transfer Status and Completion Metrics

The completed transfer status was inspected to confirm the final sender-side measurements. The transfer reached the SUCCEEDED state with the complete 262,267-byte file delivered. The recorded results included the elapsed transfer time, goodput, packet counts, acknowledgements, retransmissions, timeout events, and RTT measurements.

```
person4-impaired-demo-20260928-013805 -- zsh -- 101x29
-Dnettransfer.impairment.enabled=true \
-Dnettransfer.impairment.lossPercent=2 \
-Dnettransfer.impairment.delayMs=0 \
-Dnettransfer.impairment.seed=42 \
-Dnettransfer.impairment.scenario=loss-2pct-demo \
-Dnettransfer.logsRoot="$RUN_ROOT/logs" \
-Dnettransfer.receiverInitialTimeoutMs=240000 \
-Dnettransfer.receiverInactivityTimeoutMs=15000 \
-jar "$JAR" receiver 9000 received.bin
[impairment] loss-2pct-demo: receiver DATA loss=2.0%, DATA/ACK delivery delay=0 ms per direction, see d=42; START/FINISH unaffected
[receiver] Binding UDP socket on port 9000 ...
[receiver] Waiting for a transfer...
[receiver] Event log: /Users/mozany/Downloads/udp-file-transfer-integrated-feature-metrics-llm-integration/target/evaluation/person4-impaired-demo-20260928-013805/logs/standalone/run-lacc3268-e77a-4b29-a302-f47db34d1911/events-receiver.json1
[receiver] Saved /Users/mozany/Downloads/udp-file-transfer-integrated-feature-metrics-llm-integration/target/evaluation/person4-impaired-demo-20260928-013805/storage/incoming/received.bin
[receiver] SUCCESS: Transfer completed and verified
(base) mozany@mozas-MacBook-Air person4-impaired-demo-20260928-013805 % shasum -a 256 \
"$RUN_ROOT/storage/outgoing/sample.bin" \
"$RUN_ROOT/storage/incoming/received.bin"
873b22f9250a56ebc54af3b98f63d68b68d1287d3262a89f08ce52ce13245f4 /Users/mozany/Downloads/udp-file-transfer-integrated-feature-metrics-llm-integration/target/evaluation/person4-impaired-demo-20260928-013805/storage/outgoing/sample.bin
873b22f9250a56ebc54af3b98f63d68b68d1287d3262a89f08ce52ce13245f4 /Users/mozany/Downloads/udp-file-transfer-integrated-feature-metrics-llm-integration/target/evaluation/person4-impaired-demo-20260928-013805/storage/incoming/received.bin
(base) mozany@mozas-MacBook-Air person4-impaired-demo-20260928-013805 %

person4-FINAL-natural-language-demo -- java -Dnettransfer.impairment.enabled...
transfer> start_transfer ("file_id":"sample","receiver_id":"receiver-a","window_bytes":8192,"timeout_ms":500)
Start accepted [REAL]
transfer_id=9e9796e5-4e58-4f2a-b5c2-3b88377e035b; run_id=9e9796e5-4e58-4f2a-b5c2-3b88377e035b; protocol_transfer_id=unavailable
Requested window: 8192 bytes; effective window: 8 packets (8192 bytes); timeout: 500 ms;
retry limit: 5 consecutive rounds without progress
transfer> status last
[REAL] COMPLETED; snapshot_at=2026-09-27T21:50:51.144758Z
transfer_id=9e9796e5-4e58-4f2a-b5c2-3b88377e035b; run_id=9e9796e5-4e58-4f2a-b5c2-3b88377e035b; protocol_transfer_id=unavailable
engine_lifecycle=SUCCEEDED; evidence_scope=SENDER_ENDPOINT_LOCAL; evidence_finality=FINAL; provenance=REAL
file_size_bytes=262267
sender_confirmed_ackd_payload_bytes=262267
sender_ack_progress_percent=100
live_elapsed_time_sec=1.165878833
sender_ack_based_rate_mbps=1.7996175422459186
packets_sent=273
retransmissions=16
acks_received=257
distinct_packets_acked=257
packets_timed_out=2
rtt_sample_count=241
rtt_mean_ms=3.1231369128630706
rtt_p95_ms=7.679042
endpoint_local_udp_payload_bytes_emitted=286358
endpoint_emission_accounting_complete=true
transfer>
```

Figure 5: Receiver Completion and File Integrity Verification

The receiver reported successful transfer completion and verification. The received file was independently checked against the original source file using SHA-256, confirming that both files produced the same hash and that the transferred file was reconstructed without corruption.

```
person4-impaired-demo-20260928-013805 -- zsh -- 101x29
(base) mozany@mozas-MacBook-Air person4-FINAL-natural-language-demo % cd /Users/mozany/Downloads/udp-file-transfer-integrated-feature-metrics-llm-integration
RUN_ROOT="$PWD/target/evaluation/person4-impaired-demo-20260928-013805"
JAR="$PWD/target/udp-file-transfer.jar"
cd "$RUN_ROOT"
java \
-Dnettransfer.impairment.enabled=true \
-Dnettransfer.impairment.lossPercent=2 \
-Dnettransfer.impairment.delayMs=0 \
-Dnettransfer.impairment.seed=42 \
-Dnettransfer.impairment.scenario=loss-2pct-demo \
-Dnettransfer.logsRoot="$RUN_ROOT/logs" \
-Dnettransfer.receiverInitialTimeoutMs=240000 \
-Dnettransfer.receiverInactivityTimeoutMs=15000 \
-jar "$JAR" receiver 9000 received.bin
[impairment] loss-2pct-demo: receiver DATA loss=2.0%, DATA/ACK delivery delay=0 ms per direction, see d=42; START/FINISH unaffected
[receiver] Binding UDP socket on port 9000 ...
[receiver] Waiting for a transfer...
[receiver] Event log: /Users/mozany/Downloads/udp-file-transfer-integrated-feature-metrics-llm-integration/target/evaluation/person4-impaired-demo-20260928-013805/logs/standalone/run-lacc3268-e77a-4b29-a302-f47db34d1911/events-receiver.json1
[receiver] Saved /Users/mozany/Downloads/udp-file-transfer-integrated-feature-metrics-llm-integration/target/evaluation/person4-impaired-demo-20260928-013805/storage/incoming/received.bin
[receiver] SUCCESS: Transfer completed and verified
(base) mozany@mozas-MacBook-Air person4-impaired-demo-20260928-013805 %

person4-FINAL-natural-language-demo -- java -Dnettransfer.impairment.enabled...
transfer> start_transfer ("file_id":"sample","receiver_id":"receiver-a","window_bytes":8192,"timeout_ms":500)
Start accepted [REAL]
transfer_id=9e9796e5-4e58-4f2a-b5c2-3b88377e035b; run_id=9e9796e5-4e58-4f2a-b5c2-3b88377e035b; protocol_transfer_id=unavailable
Requested window: 8192 bytes; effective window: 8 packets (8192 bytes); timeout: 500 ms;
retry limit: 5 consecutive rounds without progress
transfer> status last
[REAL] COMPLETED; snapshot_at=2026-09-27T21:50:51.144758Z
transfer_id=9e9796e5-4e58-4f2a-b5c2-3b88377e035b; run_id=9e9796e5-4e58-4f2a-b5c2-3b88377e035b; protocol_transfer_id=unavailable
engine_lifecycle=SUCCEEDED; evidence_scope=SENDER_ENDPOINT_LOCAL; evidence_finality=FINAL; provenance=REAL
file_size_bytes=262267
sender_confirmed_ackd_payload_bytes=262267
sender_ack_progress_percent=100
live_elapsed_time_sec=1.165878833
sender_ack_based_rate_mbps=1.7996175422459186
packets_sent=273
retransmissions=16
acks_received=257
distinct_packets_acked=257
packets_timed_out=2
rtt_sample_count=241
rtt_mean_ms=3.1231369128630706
rtt_p95_ms=7.679042
endpoint_local_udp_payload_bytes_emitted=286358
endpoint_emission_accounting_complete=true
transfer>
```

Figure 6: Reconciled Transfer Evidence

The sender and receiver records were reconciled after completion to produce a final combined summary of the transfer. The reconciliation reported a COMPLETE and VERIFIED result, with successful transfer and integrity verification, and consolidated the measurements recorded by both endpoints for subsequent analysis.

```
moazany@mozas-MacBook-Air person4-FINAL-natural-language-demo % cd /Users/mozany/Downloads/udp-  
file-transfer-integrated-feature-metrics-llm-integration  
RUN_ROOT="$PWD/target/evaluation/person4-impaired-demo-20260928-013805"  
JAR="$PWD/target/udp-file-transfer.jar"  
cd "$RUN_ROOT"  
java \  
-Dnettransfer.impairment.enabled=true \  
-Dnettransfer.impairment.lossPercent=2 \  
-Dnettransfer.impairment.delayMs=0 \  
-Dnettransfer.impairment.seed=42 \  
-Dnettransfer.impairment.scenario=loss-2pct-demo \  
-Dnettransfer.logsRoot="$RUN_ROOT/logs" \  
-Dnettransfer.receiverInitialTimeoutMs=240000 \  
-Dnettransfer.receiverInactivityTimeoutMs=15000 \  
-jar "$JAR" receiver 9000 received.bin  
[impairment] loss-2pct-demo: receiver DATA loss=2.0%, DATA/ACK delivery delay=0 ms per direction, see  
d=42; START/FINISH unaffected  
[receiver] Binding UDP socket on port 9000 ...  
[receiver] Waiting for a transfer.  
[receiver] Event log: /Users/mozany/Downloads/udp-file-transfer-integrated-feature-metrics-llm-integr  
ation/target/evaluation/person4-impaired-demo-20260928-013805/logs/standalone/run-1acc3268-e77a-4b29-  
a302-f47db34d1911/events-receiver.jsonl  
[receiver] Saved /Users/mozany/Downloads/udp-file-transfer-integrated-feature-metrics-llm-integration  
/target/evaluation/person4-impaired-demo-20260928-013805/storage/incoming/received.bin  
[receiver] SUCCESS: Transfer completed and verified  
(base) moazany@mozas-MacBook-Air person4-impaired-demo-20260928-013805 %  
cd "$RUN_ROOT"  
java \  
-Dnettransfer.impairment.enabled=true \  
-Dnettransfer.impairment.lossPercent=2 \  
-Dnettransfer.impairment.delayMs=0 \  
-Dnettransfer.impairment.seed=42 \  
-Dnettransfer.impairment.scenario=loss-2pct-demo \  
-Dnettransfer.evaluation.record=true \  
-cp "$JAR" nettransfer.cli.TransferCliMain \  
"$RUN_ROOT" \  
'sample-storage/outgoing/sample.bin'  
Simulator loss-2pct-demo: receiver DATA loss=2.0%, DATA/ACK delivery delay=0 ms per dire  
ction, seed=42; START/FINISH unaffected.  
Evaluation records: /Users/mozany/Downloads/udp-file-transfer-integrated-feature-metrics  
-llm-integration/target/evaluation/person4-impaired-demo-20260928-013805/target/evaluati  
on/llm-2026-09-27T21-50-38.8569012-820932287756637956  
Recorded model responses are untrusted evidence and may include answers rejected by Java  
UDF transfer console. Type help for commands; natural-language requests use the configur  
ed interpreter.  
transfer> start_transfer ("file_id":"sample","receiver_id":"receiver-a","window_bytes":8  
192,"timeout_ms":500)  
Start accepted [REAL]  
transfer_id=9e97796e5-4e58-4f2a-b5c2-3b88377e835b; run_id=9e97796e5-4e58-4f2a-b5c2-3b88377  
e835b; protocol_transfer_id=cunavailable  
Requested window: 8192 bytes; effective window: 8 packets (8192 bytes); timeout: 500 ms;  
retry limit: 5 consecutive rounds without progress  
transfer>
```

Figure 7: Final Reconciled Metrics Summary

The reconciled evidence was inspected in greater detail to confirm the final measurements used for the demonstration. The summary contained the transfer configuration and recorded performance metrics, including delivered payload, transfer time, throughput, packet counts, retransmissions, timeout events, RTT measurements, and protocol overhead.

```
moazany@mozas-MacBook-Air person4-FINAL-natural-language-demo % cat {} \  
/Users/mozany/Downloads/udp-file-transfer-integrated-feature-metrics-llm-integration/target/evaluatio  
n/person4-FINAL-natural-language-demo/logs/standalone/reconciled/00a30b59-2c0d-445d-ad12-f33eae7b2130  
/summary.jsonl  
{  
  "schema_version": "1",  
  "metric_definition_version": "1",  
  "evidence_source": "REAL",  
  "reconciliation_status": "VERIFIED",  
  "evidence_completeness": "COMPLETE",  
  "protocol_transfer_id": "00a30b59-2c0d-445d-ad12-f33eae7b2130",  
  "sender_run_id": "23bfeca0-5d3e-488e-b366-1206b1b98253",  
  "receiver_run_id": "run-8d3c8d3b-a567-484e-98bb-5d842d7fabcc",  
  "application_transfer_id": "23bfeca0-5d3e-488e-b366-1206b1b98253",  
  "experiment_id": null,  
  "metrics": {  
    "experiment_id": null,  
    "file_size_bytes": 262267,  
    "payload_bytes_delivered": 262267,  
    "transfer_time_sec": 0.149415833,  
    "throughput_mbps": 14.042260166631738,  
    "transfer_success": true,  
    "integrity_verified": true,  
    "failure_reason": null,  
    "packets_sent": 257,  
    "packets_received": 257,  
    "packets_dropped": 0,  
    "retransmissions": 0,  
    "acks_received": 257,  
    "packets_acked": 257,  
    "packets_timed_out": 0,  
    "packets_duplicated": 0,  
    "retransmission_ratio": 0.0,  
    "udp_payload_bytes_emitted": 276729,  
    "protocol_overhead_bytes": 14462,  
    "protocol_overhead_ratio": 0.052260514799677665,  
    "rtt_sample_count": 257,  
    "rtt_mean_ms": 3.3187432178988336,  
    "rtt_p95_ms": 0.464791,  
    "chunk_size_bytes": 1024,  
    "window_bytes_requested": 8192,  
    "window_packets": 8,  
    "timeout_ms": 500,  
    "retry_limit": 5,  
    "packet_loss_rate": 0.0,  
    "delay_ms": 0.0,  
    "scenario": "final-natural-language-demo",  
    "impairment_seed": 42,  
    "impairment_mechanism": "RECEIVE_DELIVERY_V1",  
    "schema_version": "1",  
    "metric_definition_version": "1",  
    "run_id": null,  
    "application_transfer_id": "23bfeca0-5d3e-488e-b366-1206b1b98253",  
    "protocol_transfer_id": "00a30b59-2c0d-445d-ad12-f33eae7b2130",  
    "evidence_source": "REAL",  
    "endpoint_attri  
bution": "COMBINED",  
    "file_attri  
bution_timestamp": "2026-09-27T21:44:19.852928Z",  
    "finaliz  
ation_timestamp": "2026-09-27T21:44:19.852928Z",  
    "integrity_evidence_source": "RECEIVER",  
    "unavailable_reasons": {  
      "source_evidence": {  
        "sender_endpoint_record": ".../run-8d3c8d3b-a567-484e-98bb-5d842d7fabcc/events-sender.jsonl",  
        "receiver_endpoint_record": ".../run-8d3c8d3b-a567-484e-98bb-5d842d7fabcc/events-receiver.jsonl",  
        "sender_event_log": ".../run-8d3c8d3b-a567-484e-98bb-5d842d7fabcc/events-sender.jsonl",  
        "sender_run_state": ".../run-8d3c8d3b-a567-484e-98bb-5d842d7fabcc/run-state.json",  
        "receiver_run_state": ".../run-8d3c8d3b-a567-484e-98bb-5d842d7fabcc/run-state.json",  
        "finalized_at_utc": "2026-09-27T21:44:19.852928Z"  
      }  
    }  
  }  
}  
(base) moazany@mozas-MacBook-Air person4-FINAL-natural-language-demo %  
ring (for example, whether any timeout would have triggered a specific retransmission).  
Aggregate counts (packets_sent, retransmissions, packets_timed_out, acks_received, packets_acked) establish totals but do not prove event order or trigger relationships.  
Limitation: Receiver emissions or per-direction emitted-byte breakdowns are not present; udp_payload_bytes_emitted reflects successful sender socket-send UDP payload bytes as r  
econciled, and protocol_overhead_bytes is reconciled emitted minus unique receiver-deliv  
ered payload. There is no separate receiver-emitted UDP payload metric supplied.  
Limitation: Payload byte counts establish unique receiver bytes accepted (payload_bytes_  
delivered) but, by metric definitions, content integrity is established only by the sepa  
rate integrity outcome (here VERIFIED) and not by byte-count equality alone.  
Limitation: Any causal explanations (e.g., why retransmissions are zero or why RTT has t  
he observed distribution) are hypotheses only; testing them requires the additional per-  
event logs noted above (sender/receiver event traces and emulator/network configuration  
records).  
transfer> start_transfer ("file_id":"sample","receiver_id":"receiver-a","window_bytes":8  
192,"timeout_ms":1)  
INVALID_PARAMETER: timeout_ms must be between 50 and 5000  
transfer> exit  
Goodbye.  
(base) moazany@mozas-MacBook-Air person4-FINAL-natural-language-demo % unset OPENAI_API_K  
EY  
(base) moazany@mozas-MacBook-Air person4-FINAL-natural-language-demo % if [ -z "$OPENAI_A  
PI_KEY" ]; then  
  echo "API key removed from this terminal"  
else  
  echo "API key is still set"  
fi  
API key removed from this terminal  
(base) moazany@mozas-MacBook-Air person4-FINAL-natural-language-demo %
```

Figure 8: LLM Explanation Request with Recorded Evidence

After the transfer evidence had reached its final reconciled state, the LLM was asked to explain the completed transfer using only the recorded measurements. The request specifically required the explanation to report the outcome and integrity, delivered bytes, elapsed time, goodput, loss and observed drops, retransmissions, timeouts, RTT, and protocol emissions and overhead without inventing causes.

```

- exec cat {} \;
/Users/mozany/Downloads/udp-file-transfer-integrated-feature-metrics-llm-integration/target/evaluation
n/person4-FINAL-natural-language-demo/logs/standalone/reconciled/00a30b59-2c0d-445d-ad12-f33ae7b2130
/summary.jsonl
{"schema_version":"1","metric_definition_version":"1","evidence_source":"REAL","reconciliation_status":
"VERIFIED","evidence_completeness":"COMPLETE","protocol_transfer_id":"00a30b59-2c0d-445d-ad12-f33ae7b2130",
"sender_run_id":"23bfeca0-5d3e-488e-b366-1206b1b98253","receiver_run_id":"run-8d3c8d3b-a567-484e-90bb-5d842d7fabcc",
"application_transfer_id":"23bfeca0-5d3e-488e-b366-1206b1b98253","experiment_id":null,"metrics":{"experiment_id":null,"file_size_bytes":262267,"payload_bytes_delivered":262267,"t
ransfer_time_sec":0.149415833,"throughput_mbps":14.042260166631738,"transfer_success":true,"integrity_verified":true,"failure_reason":null,"packets_sent":257,"packets_received":257,"packets_dropped":0,"
retransmissions":0,"acks_received":257,"packets_acked":257,"packets_timed_out":0,"packets_duplicated":0,"retransmission_ratio":0.0,"udp_payload_bytes_emitted":276729,"protocol_overhead_bytes":14462,"pro
tocol_overhead_ratio":0.052260514799677665,"rtt_sample_count":257,"rtt_mean_ms":3.3187432178988336,"rtt_p95_ms":6.464791,"chunk_size_bytes":1024,"window_bytes_requested":8192,"window_packets":8,"timeout
_ms":1800,"retry_limit":5,"packet_loss_rate":0.0,"delay_ms":0.0,"scenario":"final-natural-language-demo",
"impairment_seed":42,"impairment_mechanism":"RECEIVE_DELIVERY_V1","schema_version":"1","metric_definition_version":"1","run_id":null,"application_transfer_id":"23bfeca0-5d3e-488e-b366-1206b1b98253","protocol_transfer_id":"00a30b59-2c0d-445d-ad12-f33ae7b2130","evidence_source":"REAL","endpoint_attri
bution":"COMBINED","file_attribution":null,"capture_timestamp":"2026-09-27T21:44:19.852928Z","finalization_timestamp":"2026-09-27T21:44:19.852928Z","integrity_evidence_source":"RECEIVER","unavailable_rea
sons":{"source_evidence":{"sender_endpoint_record":"","run-8d3c8d3b-a567-484e-90bb-5d842d7fabcc/endpoint-receiver.json","receiver_endpoint_record":"","run-8d3c8d3b-a567-484e-90bb-5d842d7fabcc/endpoint-receiver.json","sender_event_log":"","run-8d3c8d3b-a567-484e-90bb-5d842d7fabcc/events-sender.json","receiver_event_log":"","run-8d3c8d3b-a567-484e-90bb-5d842d7fabcc/events-receiver.json"},"run_state":"","run-8d3c8d3b-a567-484e-90bb-5d842d7fabcc/run-state.json","receiver_run_state":"","run-8d3c8d3b-a567-484e-90bb-5d842d7fabcc/run-state.json"},"finalized_at_utc":"2026-09-27T21:44:19.852928Z"}
(base) mozany@mozas-MacBook-Air person4-FINAL-natural-language-demo %

2d7fabcc/events-receiver.jsonl, standalone/run-8d3c8d3b-a567-484e-90bb-5d842d7fabcc/run-
state.json, standalone/reconciled/00a30b59-2c0d-445d-ad12-f33ae7b2130/summary.jsonl
Limitation: Supplied nonnumeric terminal facts: state is COMPLETED and integrity is VERI
FIED. These are reported outcomes and used here; integrity VERIFIED is the supplied rece
iver verification outcome and is not derived from byte counts.
Limitation: packets_dropped (0 packets) is the reconciled simulator receiver delivery dr
op count (definition: receiver delivery drops by RECEIVE_DELIVERY_V1). It is not a measu
red on-the-wire loss fraction and does not alone quantify all network-layer drops; packe
t_loss_rate (0.0 percent) is the configured per-arrival random-drop probability, not an
observed fraction.
Limitation: No per-packet event linkage is presented here to establish causality or orde
ring (for example, whether any timeout would have triggered a specific retransmission).
Aggregate counts (packets_sent, retransmissions, packets_timed_out, acks_received, packe
ts_acked) establish totals but do not prove event order or trigger relationships.
Limitation: Receiver emissions or per-direction emitted-byte breakdowns are not present;
udp_payload_bytes_emitted reflects successful sender socket-send UDP payload bytes as r
econciled, and protocol_overhead_bytes is reconciled emitted minus unique receiver-deliv
ered payload. There is no separate receiver-emitted UDP payload metric supplied.
Limitation: Payload byte counts establish unique receiver bytes accepted (payload_bytes_
delivered) but, by metric definitions, content integrity is established only by the sepa
rate integrity outcome (here VERIFIED) and not by byte-count equality alone.
Limitation: Any causal explanations (e.g., why retransmissions are zero or why RTT has t
he observed distribution) are hypotheses only; testing them requires the additional per-e
vent logs noted above (sender/receiver event traces and emulator/network configuration
records).
transfer> start_transfer {"file_id":"sample","receiver_id":"receiver-a","window_bytes":8
192,"seed":42,"ms":1}
INVALID PARAMETER: timeout_ms must be between 50 and 5000
transfer>
```

Figure 9: Evidence-Grounded LLM Transfer Explanation

The generated explanation used the reconciled measurements supplied by the Java application and identified the transfer as successful with verified integrity. It reported the measured transfer performance and network statistics while distinguishing recorded values from unavailable information, demonstrating that the LLM explanation was grounded in the collected evidence rather than independently generated metrics.

```

- exec cat {} \;
/Users/mozany/Downloads/udp-file-transfer-integrated-feature-metrics-llm-integration/target/evaluation
n/person4-FINAL-natural-language-demo/logs/standalone/reconciled/00a30b59-2c0d-445d-ad12-f33ae7b2130
/summary.jsonl
{"schema_version":"1","metric_definition_version":"1","evidence_source":"REAL","reconciliation_status":
"VERIFIED","evidence_completeness":"COMPLETE","protocol_transfer_id":"00a30b59-2c0d-445d-ad12-f33ae7b2130",
"sender_run_id":"23bfeca0-5d3e-488e-b366-1206b1b98253","receiver_run_id":"run-8d3c8d3b-a567-484e-90bb-5d842d7fabcc",
"application_transfer_id":"23bfeca0-5d3e-488e-b366-1206b1b98253","experiment_id":null,"metrics":{"experiment_id":null,"file_size_bytes":262267,"payload_bytes_delivered":262267,"t
ransfer_time_sec":0.149415833,"throughput_mbps":14.042260166631738,"transfer_success":true,"integrity_verified":true,"failure_reason":null,"packets_sent":257,"packets_received":257,"packets_dropped":0,"
retransmissions":0,"acks_received":257,"packets_acked":257,"packets_timed_out":0,"packets_duplicated":0,"retransmission_ratio":0.0,"udp_payload_bytes_emitted":276729,"protocol_overhead_bytes":14462,"pro
tocol_overhead_ratio":0.052260514799677665,"rtt_sample_count":257,"rtt_mean_ms":3.3187432178988336,"rtt_p95_ms":6.464791,"chunk_size_bytes":1024,"window_bytes_requested":8192,"window_packets":8,"timeout
_ms":1800,"retry_limit":5,"packet_loss_rate":0.0,"delay_ms":0.0,"scenario":"final-natural-language-demo",
"impairment_seed":42,"impairment_mechanism":"RECEIVE_DELIVERY_V1","schema_version":"1","metric_definition_version":"1","run_id":null,"application_transfer_id":"23bfeca0-5d3e-488e-b366-1206b1b98253","protocol_transfer_id":"00a30b59-2c0d-445d-ad12-f33ae7b2130","evidence_source":"REAL","endpoint_attri
bution":"COMBINED","file_attribution":null,"capture_timestamp":"2026-09-27T21:44:19.852928Z","finalization_timestamp":"2026-09-27T21:44:19.852928Z","integrity_evidence_source":"RECEIVER","unavailable_rea
sons":{"source_evidence":{"sender_endpoint_record":"","run-8d3c8d3b-a567-484e-90bb-5d842d7fabcc/endpoint-receiver.json","receiver_endpoint_record":"","run-8d3c8d3b-a567-484e-90bb-5d842d7fabcc/endpoint-receiver.json","sender_event_log":"","run-8d3c8d3b-a567-484e-90bb-5d842d7fabcc/events-sender.json","receiver_event_log":"","run-8d3c8d3b-a567-484e-90bb-5d842d7fabcc/events-receiver.json"},"run_state":"","run-8d3c8d3b-a567-484e-90bb-5d842d7fabcc/run-state.json","receiver_run_state":"","run-8d3c8d3b-a567-484e-90bb-5d842d7fabcc/run-state.json"},"finalized_at_utc":"2026-09-27T21:44:19.852928Z"}
(base) mozany@mozas-MacBook-Air person4-FINAL-natural-language-demo %

ender.json, standalone/23bfeca0-5d3e-488e-b366-1206b1b98253/events-sender.jsonl, standal
one/23bfeca0-5d3e-488e-b366-1206b1b98253/run-state.json, standalone/run-8d3c8d3b-a567-48
e-90bb-5d842d7fabcc/endpoint-receiver.json, standalone/run-8d3c8d3b-a567-484e-90bb-5d84
2d7fabcc/events-receiver.jsonl, standalone/run-8d3c8d3b-a567-484e-90bb-5d842d7fabcc/run-
state.json, standalone/reconciled/00a30b59-2c0d-445d-ad12-f33ae7b2130/summary.jsonl
Limitation: Supplied nonnumeric terminal facts: state is COMPLETED and integrity is VERI
FIED. These are reported outcomes and used here; integrity VERIFIED is the supplied rece
iver verification outcome and is not derived from byte counts.
Limitation: packets_dropped (0 packets) is the reconciled simulator receiver delivery dr
op count (definition: receiver delivery drops by RECEIVE_DELIVERY_V1). It is not a measu
red on-the-wire loss fraction and does not alone quantify all network-layer drops; packe
t_loss_rate (0.0 percent) is the configured per-arrival random-drop probability, not an
observed fraction.
Limitation: No per-packet event linkage is presented here to establish causality or orde
ring (for example, whether any timeout would have triggered a specific retransmission).
Aggregate counts (packets_sent, retransmissions, packets_timed_out, acks_received, packe
ts_acked) establish totals but do not prove event order or trigger relationships.
Limitation: Receiver emissions or per-direction emitted-byte breakdowns are not present;
udp_payload_bytes_emitted reflects successful sender socket-send UDP payload bytes as r
econciled, and protocol_overhead_bytes is reconciled emitted minus unique receiver-deliv
ered payload. There is no separate receiver-emitted UDP payload metric supplied.
Limitation: Payload byte counts establish unique receiver bytes accepted (payload_bytes_
delivered) but, by metric definitions, content integrity is established only by the sepa
rate integrity outcome (here VERIFIED) and not by byte-count equality alone.
Limitation: Any causal explanations (e.g., why retransmissions are zero or why RTT has t
he observed distribution) are hypotheses only; testing them requires the additional per-e
vent logs noted above (sender/receiver event traces and emulator/network configuration
records).
transfer>
```

Figure 10: LLM Explanation of Transfer Performance

The LLM explanation continued by reporting the recorded timing and reliability measurements from the completed run. It described the elapsed transfer time, measured goodput, packet-loss configuration, retransmission and timeout results, and RTT statistics using the values supplied in the reconciled evidence.

```
person4-FINAL-natural-language-demo -- zsh — 101x29
- exec cat {} \;
/Users/mozany/Downloads/udp-file-transfer-integrated-feature-metrics-llm-integration/target/evaluation
n/person4-FINAL-natural-language-demo/logs/standalone/reconciled/00a30b59-2c0d-445d-ad12-f33eae7b2130
/summary.jsonl
{"schema_version":"1","metric_definition_version":"1","evidence_source":"REAL","reconciliation_status
":"VERIFIED","evidence_completeness":"COMPLETE","protocol_transfer_id":"00a30b59-2c0d-445d-ad12-f33eae7b2130","sender_run_id":"23bfeca0-5d3e-488e-b366-1206b1b98253","receiver_run_id":"run-8d3c8d3b-a567-484e-90bb-5d842d7fabcc","application_transfer_id":"23bfeca0-5d3e-488e-b366-1206b1b98253","experiment_id":null,"metrics":{"experiment_id":null,"file_size_bytes":262267,"payload_bytes_delivered":262267,"rt_transfer_time_sec":0.149415833,"throughput_mbps":14.042260166631738,"transfer_success":true,"integrity_verified":true,"failure_reason":null,"packets_sent":257,"packets_received":257,"packets_dropped":0,"retransmissions":0,"acks_received":257,"packets_acked":257,"packets_timed_out":0,"packets_duplicated":0,"retransmission_ratio":0.0,"udp_payload_bytes_emitted":276729,"protocol_overhead_bytes":14462,"protocol_overhead_ratio":0.052260514799677665,"rtt_sample_count":257,"rtt_mean_ms":3.3187432178988336,"rtt_p95_ms":6.464791,"chunk_size_bytes":1024,"window_bytes_requested":8192,"window_packets":8,"timeout_ms":500,"retry_limit":5,"packet_loss_rate":0.0,"delay_ms":0.0,"scenario":"final-natural-language-demo","impairment_seed":42,"impairment_mechanism":"RECEIVE_DELIVERY_V1","schema_version":"1","metric_definition_version":"1","run_id":null,"application_transfer_id":"23bfeca0-5d3e-488e-b366-1206b1b98253","protocol_transfer_id":"00a30b59-2c0d-445d-ad12-f33eae7b2130","evidence_source":"REAL","endpoint_attribution":"COMBINED","file_attribution":null,"capture_timestamp":"2026-09-27T21:44:19.852928Z","finalization_timestamp":"2026-09-27T21:44:19.852928Z","integrity_evidence_source":"RECEIVER","unavailable_reasons":{},"source_evidence":{"sender_endpoint_record":".../23bfeca0-5d3e-488e-b366-1206b1b98253/endpoint-sender.json","receiver_endpoint_record":".../run-8d3c8d3b-a567-484e-90bb-5d842d7fabcc/endpoint-receiver.json","sender_event_log":".../23bfeca0-5d3e-488e-b366-1206b1b98253/events-sender.json","receiver_event_log":".../run-8d3c8d3b-a567-484e-90bb-5d842d7fabcc/events-receiver.json"},"sender_run_state":".../23bfeca0-5d3e-488e-b366-1206b1b98253/run-state.json","receiver_run_state":".../run-8d3c8d3b-a567-484e-90bb-5d842d7fabcc/run-state.json"},"finalized_at_utc":"2026-09-27T21:44:19.852928Z"}
(base) mozany@mozas-MacBook-Air person4-FINAL-natural-language-demo %
```

```
person4-FINAL-natural-language-demo -- java -Dnettransfer.impairment.enabled...
ender.json, standalone/23bfeca0-5d3e-488e-b366-1206b1b98253/events-sender.json, standalone/23bfeca0-5d3e-488e-b366-1206b1b98253/run-state.json, standalone/run-8d3c8d3b-a567-484e-90bb-5d842d7fabcc/endpoint-receiver.json, standalone/run-8d3c8d3b-a567-484e-90bb-5d842d7fabcc/events-receiver.json, standalone/reconciled/00a30b59-2c0d-445d-ad12-f33eae7b2130/summary.jsonl. These are reported outcomes and used here; integrity VERIFIED is the supplied receiver verification outcome and is not derived from byte counts.
Limitation: packets_dropped (0 packets) is the reconciled simulator receiver delivery drop count (definition: receiver delivery drops by RECEIVE_DELIVERY_V1). It is not a measured on-the-wire loss fraction and does not alone quantify all network-layer drops; packet_loss_rate (0.0 percent) is the configured per-arrival random-drop probability, not an observed fraction.
Limitation: No per-packet event linkage is presented here to establish causality or ordering (for example, whether any timeout would have triggered a specific retransmission). Aggregate counts (packets_sent, retransmissions, packets_timed_out, acks_received, packets_acked) establish totals but do not prove event order or trigger relationships.
Limitation: Receiver emissions or per-direction emitted-byte breakdowns are not present; udp_payload_bytes_emitted reflects successful sender socket-send UDP payload bytes as reconciled, and protocol_overhead_bytes is reconciled emitted minus unique receiver-delivered payload. There is no separate receiver-emitted UDP payload metric supplied.
Limitation: Payload byte counts establish unique receiver bytes accepted (payload_bytes_delivered) but, by metric definitions, content integrity is established only by the separate integrity outcome (here VERIFIED) and not by byte-count equality alone.
Limitation: Any causal explanations (e.g., why retransmissions are zero or why RTT has the observed distribution) are hypotheses only; testing them requires the additional per-event logs noted above (sender/receiver event traces and emulator/network configuration records).
transfer>
```

Figure 11: LLM Explanation of Protocol Overhead

The explanation also reported the combined sender-and-receiver UDP payload emissions and the resulting protocol overhead. These values were taken from the reconciled transfer summary, allowing the LLM to describe communication overhead using recorded measurements rather than estimating or generating values independently.

```
person4-FINAL-natural-language-demo -- zsh — 101x29
- exec cat {} \;
/Users/mozany/Downloads/udp-file-transfer-integrated-feature-metrics-llm-integration/target/evaluation
n/person4-FINAL-natural-language-demo/logs/standalone/reconciled/00a30b59-2c0d-445d-ad12-f33eae7b2130
/summary.jsonl
{"schema_version":"1","metric_definition_version":"1","evidence_source":"REAL","reconciliation_status
":"VERIFIED","evidence_completeness":"COMPLETE","protocol_transfer_id":"00a30b59-2c0d-445d-ad12-f33eae7b2130","sender_run_id":"23bfeca0-5d3e-488e-b366-1206b1b98253","receiver_run_id":"run-8d3c8d3b-a567-484e-90bb-5d842d7fabcc","application_transfer_id":"23bfeca0-5d3e-488e-b366-1206b1b98253","experiment_id":null,"metrics":{"experiment_id":null,"file_size_bytes":262267,"payload_bytes_delivered":262267,"rt_transfer_time_sec":0.149415833,"throughput_mbps":14.042260166631738,"transfer_success":true,"integrity_verified":true,"failure_reason":null,"packets_sent":257,"packets_received":257,"packets_dropped":0,"retransmissions":0,"acks_received":257,"packets_acked":257,"packets_timed_out":0,"packets_duplicated":0,"retransmission_ratio":0.0,"udp_payload_bytes_emitted":276729,"protocol_overhead_bytes":14462,"protocol_overhead_ratio":0.052260514799677665,"rtt_sample_count":257,"rtt_mean_ms":3.3187432178988336,"rtt_p95_ms":6.464791,"chunk_size_bytes":1024,"window_bytes_requested":8192,"window_packets":8,"timeout_ms":500,"retry_limit":5,"packet_loss_rate":0.0,"delay_ms":0.0,"scenario":"final-natural-language-demo","impairment_seed":42,"impairment_mechanism":"RECEIVE_DELIVERY_V1","schema_version":"1","metric_definition_version":"1","run_id":null,"application_transfer_id":"23bfeca0-5d3e-488e-b366-1206b1b98253","protocol_transfer_id":"00a30b59-2c0d-445d-ad12-f33eae7b2130","evidence_source":"REAL","endpoint_attribution":"COMBINED","file_attribution":null,"capture_timestamp":"2026-09-27T21:44:19.852928Z","finalization_timestamp":"2026-09-27T21:44:19.852928Z","integrity_evidence_source":"RECEIVER","unavailable_reasons":{},"source_evidence":{"sender_endpoint_record":".../23bfeca0-5d3e-488e-b366-1206b1b98253/endpoint-sender.json","receiver_endpoint_record":".../run-8d3c8d3b-a567-484e-90bb-5d842d7fabcc/endpoint-receiver.json","sender_event_log":".../23bfeca0-5d3e-488e-b366-1206b1b98253/events-sender.json","receiver_event_log":".../run-8d3c8d3b-a567-484e-90bb-5d842d7fabcc/events-receiver.json"},"sender_run_state":".../23bfeca0-5d3e-488e-b366-1206b1b98253/run-state.json","receiver_run_state":".../run-8d3c8d3b-a567-484e-90bb-5d842d7fabcc/run-state.json"},"finalized_at_utc":"2026-09-27T21:44:19.852928Z"}
(base) mozany@mozas-MacBook-Air person4-FINAL-natural-language-demo %
```

```
person4-FINAL-natural-language-demo -- java -Dnettransfer.impairment.enabled...
transfer> ask Send sample to receiver-a using an 8192-byte window and a 500-millisecond
Start accepted [REAL]
transfer_id=23bfeca0-5d3e-488e-b366-1206b1b98253; run_id=23bfeca0-5d3e-488e-b366-1206b1b98253; protocol_transfer_id=unavailable
Requested window: 8192 bytes; effective window: 8 packets (8192 bytes); timeout: 500 ms;
retry limit: 5 consecutive rounds without progress
transfer> status last
[REAL] COMPLETED; snapshot_at=2026-09-27T21:41:19.555526Z
transfer_id=23bfeca0-5d3e-488e-b366-1206b1b98253; run_id=23bfeca0-5d3e-488e-b366-1206b1b98253; protocol_transfer_id=00a30b59-2c0d-445d-ad12-f33eae7b2130
engine_lifecycle=SUCCEEDED; evidence_scope=SENDER_ENDPOINT_LOCAL; evidence_finality=FINA
l; provenance=REAL
file_size_bytes=262267
sender_confirmed_acked_payload_bytes=262267
sender_ack_progress_percent=100
live_elapsed_time_sec=0.149415833
sender_ack_based_rate_mbps=14.042260166631738
packets_sent=257
retransmissions=0
acks_received=257
distinct_packets_acked=257
packets_timed_out=0
rtt_sample_count=257
rtt_mean_ms=3.3187432178988336
rtt_p95_ms=6.464791
endpoint_local_udp_payload_bytes_emitted=269542
endpoint_emission_accounting_complete=true
transfer>
```

Figure 12: Completion of the Evidence-Grounded Explanation

The LLM completed its response while remaining within the available recorded evidence and explicitly identifying information that was not available where applicable. Control then returned to the transfer console, confirming completion of the explanation stage of the demonstration.

```
person4-FINAL-natural-language-demo -- zsh — 101x29
mo/storage/incoming/received.bin
(base) mozany@mozas-MacBook-Air person4-FINAL-natural-language-demo % SENDER_DIR=$(dirname "$(find "$RUN_ROOT/logs/standalone" -name endpoint-sender.json -type f | head -1)")
RECEIVER_DIR=$(dirname "$(find "$RUN_ROOT/logs/standalone" -name endpoint-receiver.json -type f | head -1)")

echo "SENDER=$SENDER_DIR"
echo "RECEIVER=$RECEIVER_DIR"
SENDER=/Users/mozany/Downloads/udp-file-transfer-integrated-feature-metrics-llm-integration/target/evaluation/person4-FINAL-natural-language-demo/logs/standalone/23bfeca0-5d3e-488e-b366-1206b1b98253
RECEIVER=/Users/mozany/Downloads/udp-file-transfer-integrated-feature-metrics-llm-integration/target/evaluation/person4-FINAL-natural-language-demo/logs/standalone/run-8d3c8d3b-a567-484e-90bb-5d842d7fabcc
(base) mozany@mozas-MacBook-Air person4-FINAL-natural-language-demo % java -jar "$JAR" reconcile "$SENDER_DIR" "$RECEIVER_DIR"
WARNING: Final field chunkSizeBytes in class nettransfer.metrics.TransferConfiguration has been mutated reflectively by class com.google.gson.internal.bind.ReflectiveTypeAdapterFactory$2 in unnamed module @7b6cd12c (file:/Users/mozany/Downloads/udp-file-transfer-integrated-feature-metrics-llm-integration/target/udp-file-transfer.jar)
WARNING: Use --enable-final-field-mutation=ALL-UNNAMED to avoid a warning
WARNING: Mutating final fields will be blocked in a future release unless final field mutation is enabled
[metrics] Summary: /Users/mozany/Downloads/udp-file-transfer-integrated-feature-metrics-llm-integration/target/evaluation/person4-FINAL-natural-language-demo/logs/standalone/reconciled/00a30b59-2c0d-445d-ad12-f33eae7b2130/summary.jsonl
[metrics] Manifest: /Users/mozany/Downloads/udp-file-transfer-integrated-feature-metrics-llm-integration/target/evaluation/person4-FINAL-natural-language-demo/logs/standalone/reconciled/00a30b59-2c0d-445d-ad12-f33eae7b2130/manifest.json
(base) mozany@mozas-MacBook-Air person4-FINAL-natural-language-demo %
```

```
person4-FINAL-natural-language-demo -- java -Dnettransfer.impairment.enabled...
transfer> ask Send sample to receiver-a using an 8192-byte window and a 500-millisecond timeout.
Start accepted [REAL]
transfer_id=23bfeca0-5d3e-488e-b366-1206b1b98253; run_id=23bfeca0-5d3e-488e-b366-1206b1b98253; protocol_transfer_id=unavailable
Requested window: 8192 bytes; effective window: 8 packets (8192 bytes); timeout: 500 ms;
retry limit: 5 consecutive rounds without progress
transfer> status last
[REAL] COMPLETED; snapshot_at=2026-09-27T21:41:19.555526Z
transfer_id=23bfeca0-5d3e-488e-b366-1206b1b98253; run_id=23bfeca0-5d3e-488e-b366-1206b1b98253; protocol_transfer_id=unavailable
engine_lifecycle=SUCCEEDED; evidence_scope=SENDER_ENDPOINT_LOCAL; evidence_finality=FINAL; provenance=REAL
file_size_bytes=262267
sender_confirmed_acked_payload_bytes=262267
sender_ack_progress_percent=100
live_elapsed_time_sec=0.149615833
sender_ack_based_rate_mbps=14.042260166631738
packets_sent=257
retransmissions=0
acks_received=257
distinct_packets_acked=257
packets_timed_out=0
rtt_sample_count=257
rtt_mean_ms=3.3187432178988336
rtt_p95_ms=6.464791
endpoint_local_udp_payload_bytes_emitted=269542
endpoint_emission_accounting_complete=true
transfer>
```

Figure 13: Deterministic Invalid-Command Test

After completing the successful natural-language demonstration, an intentionally invalid structured transfer command was submitted with a timeout value of 1 ms. This value was deliberately selected outside the range permitted by the Java validation layer to test deterministic rejection of unsafe or invalid parameters.

```
person4-FINAL-natural-language-demo -- zsh — 101x29
[receiver] Waiting for a transfer...
[receiver] Event log: /Users/mozany/Downloads/udp-file-transfer-integrated-feature-metrics-llm-integration/target/evaluation/person4-FINAL-natural-language-demo/logs/standalone/run-8d3c8d3b-a567-484e-90bb-5d842d7fabcc/events-receiver.jsonl
[receiver] Saved /Users/mozany/Downloads/udp-file-transfer-integrated-feature-metrics-llm-integration/target/evaluation/person4-FINAL-natural-language-demo/storage/incoming/received.bin
[receiver] SUCCESS: Transfer completed and verified
(base) mozany@mozas-MacBook-Air person4-FINAL-natural-language-demo % shasum -a 256 \
"$RUN_ROOT/storage/outgoing/sample.bin" \
"$RUN_ROOT/storage/incoming/received.bin"
873b22f9250a56ebc54af3b9f63d680608d1207d3262a89f90ce52ce13245f4 /Users/mozany/Downloads/udp-file-transfer-integrated-feature-metrics-llm-integration/target/evaluation/person4-FINAL-natural-language-demo/storage/outgoing/sample.bin
873b22f9250a56ebc54af3b9f63d680608d1207d3262a89f90ce52ce13245f4 /Users/mozany/Downloads/udp-file-transfer-integrated-feature-metrics-llm-integration/target/evaluation/person4-FINAL-natural-language-demo/storage/incoming/received.bin
(base) mozany@mozas-MacBook-Air person4-FINAL-natural-language-demo % SENDER_DIR=$(dirname "$(find "$RUN_ROOT/logs/standalone" -name endpoint-sender.json -type f | head -1)")
RECEIVER_DIR=$(dirname "$(find "$RUN_ROOT/logs/standalone" -name endpoint-receiver.json -type f | head -1)")

echo "SENDER=$SENDER_DIR"
echo "RECEIVER=$RECEIVER_DIR"
SENDER=/Users/mozany/Downloads/udp-file-transfer-integrated-feature-metrics-llm-integration/target/evaluation/person4-FINAL-natural-language-demo/logs/standalone/23bfeca0-5d3e-488e-b366-1206b1b98253
RECEIVER=/Users/mozany/Downloads/udp-file-transfer-integrated-feature-metrics-llm-integration/target/evaluation/person4-FINAL-natural-language-demo/logs/standalone/run-8d3c8d3b-a567-484e-90bb-5d842d7fabcc
(base) mozany@mozas-MacBook-Air person4-FINAL-natural-language-demo %
```

```
person4-FINAL-natural-language-demo -- java -Dnettransfer.impairment.enabled...
transfer> ask Send sample to receiver-a using an 8192-byte window and a 500-millisecond timeout.
Start accepted [REAL]
transfer_id=23bfeca0-5d3e-488e-b366-1206b1b98253; run_id=23bfeca0-5d3e-488e-b366-1206b1b98253; protocol_transfer_id=unavailable
Requested window: 8192 bytes; effective window: 8 packets (8192 bytes); timeout: 500 ms;
retry limit: 5 consecutive rounds without progress
transfer> status last
[REAL] COMPLETED; snapshot_at=2026-09-27T21:41:19.555526Z
transfer_id=23bfeca0-5d3e-488e-b366-1206b1b98253; run_id=23bfeca0-5d3e-488e-b366-1206b1b98253; protocol_transfer_id=00a30b59-2c0d-445d-ad12-f33eae7b2130
engine_lifecycle=SUCCEEDED; evidence_scope=SENDER_ENDPOINT_LOCAL; evidence_finality=FINAL; provenance=REAL
file_size_bytes=262267
sender_confirmed_acked_payload_bytes=262267
sender_ack_progress_percent=100
live_elapsed_time_sec=0.149615833
sender_ack_based_rate_mbps=14.042260166631738
packets_sent=257
retransmissions=0
acks_received=257
distinct_packets_acked=257
packets_timed_out=0
rtt_sample_count=257
rtt_mean_ms=3.3187432178988336
rtt_p95_ms=6.464791
endpoint_local_udp_payload_bytes_emitted=269542
endpoint_emission_accounting_complete=true
transfer>
```

Figure 14: Invalid Parameter Rejection

The invalid transfer request was rejected by the Java-controlled interface with an INVALID_PARAMETER response because the requested 1 ms timeout was below the permitted range of 50–5000 ms. No new transfer was started, demonstrating that execution remained controlled by deterministic Java validation rather than by the LLM.

```
person4-FINAL-natural-language-demo -- zsh — 101x29
-Dnettransfer.impairment.enabled=true \
-Dnettransfer.impairment.lossPercent=0 \
-Dnettransfer.impairment.delayMs=0 \
-Dnettransfer.impairment.seed=42 \
-Dnettransfer.impairment.scenario=final-natural-language-demo \
-Dnettransfer.logsRoot="$RUN_ROOT/logs" \
-Dnettransfer.receiverInitialTimeoutMs=240000 \
-Dnettransfer.receiverInactivityTimeoutMs=15000 \
-jar "$JAR" receiver 9000 received.bin
[impairment] final-natural-language-demo: receiver DATA loss=0.0%, DATA/ACK delivery delay=0 ms per d
irection, seed=42; START/FINISH unaffected
[receiver] Binding UDP socket on port 9000 ...
[receiver] Waiting for a transfer...
[receiver] Event log: /Users/mozany/Downloads/udp-file-transfer-integrated-feature-metrics-llm-integr
ation/target/evaluation/person4-FINAL-natural-language-demo/logs/standalone/run-8d3c8d3b-a567-484e-9b
bb-5d842d7fabcc/events-receiver.jsonl
[receiver] Saved /Users/mozany/Downloads/udp-file-transfer-integrated-feature-metrics-llm-integration
/target/evaluation/person4-FINAL-natural-language-demo/storage/incoming/received.bin
[receiver] SUCCESS: Transfer completed and verified
(base) mozany@mozas-MacBook-Air person4-FINAL-natural-language-demo % shasum -a 256 \
"$RUN_ROOT/storage/outgoing/sample.bin" \
873b22f9250a5656bc54f3b9f8f63d68808d1207d3262a89f08ce52ce13245f4 /Users/mozany/Downloads/udp-file-tr
ansfer-integrated-feature-metrics-llm-integration/target/evaluation/person4-FINAL-natural-language-de
mo/storage/outgoing/sample.bin
873b22f9250a5656bc54f3b9f8f63d68808d1207d3262a89f08ce52ce13245f4 /Users/mozany/Downloads/udp-file-tr
ansfer-integrated-feature-metrics-llm-integration/target/evaluation/person4-FINAL-natural-language-de
mo/storage/incoming/received.bin
(base) mozany@mozas-MacBook-Air person4-FINAL-natural-language-demo %
```

```
person4-FINAL-natural-language-demo -- java -Dnettransfer.impairment.enabled...
transfer> ask Send sample to receiver-a using an 8192-byte window and a 500-millisecond
timeout.
Start accepted [REAL]
transfer_id=23bfeca0-5d3e-488e-b366-1206b1b98253; run_id=23bfeca0-5d3e-488e-b366-1206b1b
98253; protocol_transfer_id=unavailable
Requested window: 8192 bytes; effective window: 8 packets (8192 bytes); timeout: 500 ms;
retry limit: 5 consecutive rounds without progress
transfer> status last
[REAL] COMPLETED; snapshot_at=2026-09-27T21:41:19.555526Z
transfer_id=23bfeca0-5d3e-488e-b366-1206b1b98253; run_id=23bfeca0-5d3e-488e-b366-1206b1b
98253; protocol_transfer_id=00a30b59-2c8d-445d-ad12-f33ae7b2130
engine_lifecycle=SUCCEEDED; evidence_scope=SENDER_ENDPOINT_LOCAL; evidence_finality=FINA
L; provenance=REAL
file_size_bytes=262267
sender_confirmed_acked_payload_bytes=262267
sender_ack_progress_percent=100
live_elapsed_time_sec=0.149415833
sender_ack_based_rate_mbps=14.042260166631738
packets_sent=257
retransmissions=0
acks_received=257
distinct_packets_acked=257
packets_timed_out=0
rtt_sample_count=257
rtt_mean_ms=3.3187432178988336
rtt_p95_ms=6.464791
endpoint_local_udp_payload_bytes_emitted=269542
endpoint_emission_accounting_complete=true
transfer>
```

Figure 15: Completion of the Natural-Language Demonstration

Following the invalid-command test, the console session was terminated normally. The completed demonstration therefore provided evidence of a successful natural-language transfer, verified file integrity, an LLM explanation grounded in recorded measurements, and deterministic rejection of an invalid transfer request.

```
person4-FINAL-natural-language-demo -- zsh — 101x29
FINAL RUN: /Users/mozany/Downloads/udp-file-transfer-integrated-feature-metrics-llm-integration/targe
t/evaluation/person4-FINAL-natural-language-demo
873b22f9250a5656bc54f3b9f8f63d68808d1207d3262a89f08ce52ce13245f4 /Users/mozany/Downloads/udp-file-tr
ansfer-integrated-feature-metrics-llm-integration/target/evaluation/person4-FINAL-natural-language-de
mo/storage/outgoing/sample.bin
(base) mozany@mozas-MacBook-Air udp-file-transfer-integrated-feature-metrics-llm-integration % cd "$R
UN_ROOT"

java \
-Dnettransfer.impairment.enabled=true \
-Dnettransfer.impairment.lossPercent=0 \
-Dnettransfer.impairment.delayMs=0 \
-Dnettransfer.impairment.seed=42 \
-Dnettransfer.impairment.scenario=final-natural-language-demo \
-Dnettransfer.logsRoot="$RUN_ROOT/logs" \
-Dnettransfer.receiverInitialTimeoutMs=240000 \
-Dnettransfer.receiverInactivityTimeoutMs=15000 \
-jar "$JAR" receiver 9000 received.bin
[impairment] final-natural-language-demo: receiver DATA loss=0.0%, DATA/ACK delivery delay=0 ms per d
irection, seed=42; START/FINISH unaffected
[receiver] Binding UDP socket on port 9000 ...
[receiver] Waiting for a transfer...
[receiver] Event log: /Users/mozany/Downloads/udp-file-transfer-integrated-feature-metrics-llm-integr
ation/target/evaluation/person4-FINAL-natural-language-demo/logs/standalone/run-8d3c8d3b-a567-484e-9b
bb-5d842d7fabcc/events-receiver.jsonl
[receiver] Saved /Users/mozany/Downloads/udp-file-transfer-integrated-feature-metrics-llm-integration
/target/evaluation/person4-FINAL-natural-language-demo/storage/incoming/received.bin
[receiver] SUCCESS: Transfer completed and verified
(base) mozany@mozas-MacBook-Air person4-FINAL-natural-language-demo %
```

```
person4-FINAL-natural-language-demo -- java -Dnettransfer.impairment.enabled...
transfer> ask Send sample to receiver-a using an 8192-byte window and a 500-millisecond
timeout.
Start accepted [REAL]
transfer_id=23bfeca0-5d3e-488e-b366-1206b1b98253; run_id=23bfeca0-5d3e-488e-b366-1206b1b
98253; protocol_transfer_id=unavailable
Requested window: 8192 bytes; effective window: 8 packets (8192 bytes); timeout: 500 ms;
retry limit: 5 consecutive rounds without progress
transfer> status last
[REAL] COMPLETED; snapshot_at=2026-09-27T21:41:19.555526Z
transfer_id=23bfeca0-5d3e-488e-b366-1206b1b98253; run_id=23bfeca0-5d3e-488e-b366-1206b1b
98253; protocol_transfer_id=00a30b59-2c8d-445d-ad12-f33ae7b2130
engine_lifecycle=SUCCEEDED; evidence_scope=SENDER_ENDPOINT_LOCAL; evidence_finality=FINA
L; provenance=REAL
file_size_bytes=262267
sender_confirmed_acked_payload_bytes=262267
sender_ack_progress_percent=100
live_elapsed_time_sec=0.149415833
sender_ack_based_rate_mbps=14.042260166631738
packets_sent=257
retransmissions=0
acks_received=257
distinct_packets_acked=257
packets_timed_out=0
rtt_sample_count=257
rtt_mean_ms=3.3187432178988336
rtt_p95_ms=6.464791
endpoint_local_udp_payload_bytes_emitted=269542
endpoint_emission_accounting_complete=true
transfer>
```

Figure 16: Impaired Transfer Environment Setup

A separate impaired demonstration was then prepared using the same small test file. The network impairment simulator was enabled with a configured 2% DATA packet-loss rate, zero added delay, and a fixed seed of 42 to provide a controlled and reproducible loss scenario.

```

person4-FINAL-natural-language-demo -- zsh -- 101x29
FINAL RUN: /Users/mozany/Downloads/udp-file-transfer-integrated-feature-metrics-llm-integration/target/evaluation/person4-FINAL-natural-language-demo
873b22f9250e65ebc54f3b9f6d3d68808d120d33262a89f08ce52ce13245f4 /Users/mozany/Downloads/udp-file-transfer-integrated-feature-metrics-llm-integration/target/evaluation/person4-FINAL-natural-language-demo/storage/outgoing/sample.bin
(base) mozany@mozas-MacBook-Air udp-file-transfer-integrated-feature-metrics-llm-integration % cd "$RUN_ROOT"
java \
-Dnettransfer.impairment.enabled=true \
-Dnettransfer.impairment.lossPercent=0 \
-Dnettransfer.impairment.delayMs=0 \
-Dnettransfer.impairment.seed=42 \
-Dnettransfer.impairment.scenario=final-natural-language-demo \
-Dnettransfer.logsRoot="$RUN_ROOT/logs" \
-Dnettransfer.receiverInitialTimeoutMs=248000 \
-Dnettransfer.receiverInactivityTimeoutMs=15000 \
-jar "$JAR" receiver 9800 received.bin
[impairment] final-natural-language-demo: receiver DATA loss=0.0%, DATA/ACK delivery delay=0 ms per direction, seed=42; START/FINISH unaffected
[receiver] Binding UDP socket on port 9800 ...
[receiver] Waiting for a transfer...
[receiver] Event log: /Users/mozany/Downloads/udp-file-transfer-integrated-feature-metrics-llm-integration/target/evaluation/person4-FINAL-natural-language-demo/logs/standalone/run-8d3c8d3b-a567-484e-90bb-5d842d7fabcc/events-receiver.jsonl
[receiver] Saved /Users/mozany/Downloads/udp-file-transfer-integrated-feature-metrics-llm-integration/target/evaluation/person4-FINAL-natural-language-demo/storage/incoming/received.bin
[receiver] SUCCESS: Transfer completed and verified
(base) mozany@mozas-MacBook-Air person4-FINAL-natural-language-demo %

person4-FINAL-natural-language-demo -- java -Dnettransfer.impairment.enabled...
API key is set
(base) mozany@mozas-MacBook-Air person4-FINAL-natural-language-demo % java \
-Dnettransfer.impairment.enabled=true \
-Dnettransfer.impairment.lossPercent=0 \
-Dnettransfer.impairment.delayMs=0 \
-Dnettransfer.impairment.seed=42 \
-Dnettransfer.impairment.scenario=final-natural-language-demo \
-Dnettransfer.evaluation.record=true \
-cp "$JAR" nettransfer.cli.TransferCliMain \
"$RUN_ROOT" \
'sample-storage/outgoing/sample.bin'
Simulator final-natural-language-demo: receiver DATA loss=0.0%, DATA/ACK delivery delay=0 ms per direction, seed=42; START/FINISH unaffected.
Evaluation records: /Users/mozany/Downloads/udp-file-transfer-integrated-feature-metrics-llm-integration/target/evaluation/person4-FINAL-natural-language-demo/target/evaluation/llm-2026-09-27T21-41-03.3636852-17848341343288940225
Recorded model responses are untrusted evidence and may include answers rejected by Java
UDP transfer console. Type help for commands; natural-language requests use the configured interpreter.
transfer> ask Send sample to receiver-a using an 8192-byte window and a 500-millisecond timeout.
Start accepted [REAL]
transfer_id=23bfeca8-5d3e-488e-b366-1206b1b98253; run_id=23bfeca8-5d3e-488e-b366-1206b1b98253; protocol_transfer_id=unavailable
Requested window: 8192 bytes; effective window: 8 packets (8192 bytes); timeout: 500 ms;
retry limit: 5 consecutive rounds without progress
transfer>

```

Figure 17: Impaired Transfer Initiation

The impaired transfer was started using an 8192-byte window and a 500 ms timeout. The Java-controlled console accepted the request as a REAL transfer, corresponding to an effective sliding window of eight packets while the receiver operated under the configured 2% loss impairment.

```

person4-natural-language-success-20260928-011843 -- zsh -- 101x29
/Users/mozany/Downloads/udp-file-transfer-integrated-feature-metrics-llm-integration/target/evaluation/person4-natural-language-success-20260928-011843/logs/standalone/reconciled/d07175fd-d5c6-46c1-8980-1b314300aeb9/summary.jsonl
{"schema_version":"1.1","metric_definition_version":"1.1","evidence_source":"REAL","reconciliation_status":"VERIFIED","evidence_completeness":"COMPLETE","protocol_transfer_id":"d07175fd-d5c6-46c1-8980-1b314300aeb9","sender_run_id":"9849d0e3-4ae5-4e7f-bb29-262e97812440","receiver_run_id":"run-6c14faa5-7593-44e5-a354-a71d9ddf2e65","application_transfer_id":"9849d0e3-4ae5-4e7f-bb29-262e97812440","experiment_id":null,"metrics":{"experiment_id":null,"file_size_bytes":262267,"payload_bytes_delivered":262267,"transfer_time_sec":0.117496667,"throughput_mbps":17.856983126168167,"transfer_success":true,"integrity_verified":true,"failure_reason":null,"packets_sent":257,"packets_received":257,"packets_dropped":0,"retransmissions":0,"acks_received":257,"packets_acked":257,"packets_timed_out":0,"packets_duplicated":0,"retransmission_ratio":0.0,"udp_payload_bytes_emitted":276729,"protocol_overhead_bytes":14462,"protocol_overhead_ratio":0.052260514799677665,"rtt_sample_count":257,"rtt_mean_ms":2.5661569610894963,"rtt_p95_ms":6.881125,"chunk_size_bytes":1024,"window_bytes_requested":8192,"window_packets":8,"timeout_ms":500,"retry_limit":5,"packet_loss_rate":0.0,"delay_ms":0.0,"scenario":"natural-language-success","impairment_seed":42,"impairment_mechanism":"RECEIVE_DELIVERY_V2","schema_version":"1.1","metric_definition_version":"1.1","run_id":null,"application_transfer_id":"9849d0e3-4ae5-4e7f-bb29-262e97812440","protocol_transfer_id":"d07175fd-d5c6-46c1-8980-1b314300aeb9","evidence_source":"REAL","endpoint_attribution":"COMBINED","file_attribution":null,"capture_timestamp":"2026-09-27T21:24:09.1860992","finalization_timestamp":"2026-09-27T21:24:09.1860992","integrity_evidence_source":"RECEIVER","unavailable_reason":{"source_evidence":{"sender_endpoint_record":"./run-9849d0e3-4ae5-4e7f-bb29-262e97812440/endpoint-sender.json","receiver_endpoint_record":"./run-6c14faa5-7593-44e5-a354-a71d9ddf2e65/endpoint-receiver.json","sender_event_log":"./run-9849d0e3-4ae5-4e7f-bb29-262e97812440/events-sender.jsonl","receiver_event_log":"./run-6c14faa5-7593-44e5-a354-a71d9ddf2e65/events-receiver.jsonl","sender_run_state":"./run-9849d0e3-4ae5-4e7f-bb29-262e97812440/run-state.json","receiver_run_state":"./run-6c14faa5-7593-44e5-a354-a71d9ddf2e65/run-state.json"},"finalized_at_utc":"2026-09-27T21:24:09.1860992"}}
(base) mozany@mozas-MacBook-Air person4-natural-language-success-20260928-011843 %

person4-natural-language-success-20260928-011843 -- zsh -- 88x28
-Dnettransfer.impairment.enabled=true \
-Dnettransfer.impairment.lossPercent=0 \
-Dnettransfer.impairment.delayMs=0 \
-Dnettransfer.impairment.seed=42 \
-Dnettransfer.impairment.scenario=natural-language-success \
-Dnettransfer.evaluation.record=true \
-cp "$JAR" nettransfer.cli.TransferCliMain \
"$RUN_ROOT" \
'sample-storage/outgoing/sample.bin'
Simulator natural-language-success: receiver DATA loss=0.0%, DATA/ACK delivery delay=0 ms per direction, seed=42; START/FINISH unaffected.
Evaluation records: /Users/mozany/Downloads/udp-file-transfer-integrated-feature-metrics-llm-integration/target/evaluation/person4-natural-language-success-20260928-011843/target/evaluation/llm-2026-09-27T21-36-53.8065592-13862450360272930422
Recorded model responses are untrusted evidence and may include answers rejected by Java
UDP transfer console. Type help for commands; natural-language requests use the configured interpreter.
transfer> explain {"run_id":"9849d0e3-4ae5-4e7f-bb29-262e97812440","question":"Summarize this run using only the supplied measurements: outcome and integrity, delivered bytes, transfer time and goodput, configured loss versus observed simulator drops, retransmissions and timeouts, RTT, and combined sender-and-receiver UDP payload emissions and overhead. State unavailable values and uncertainty; do not invent causes or recommendations."}
UNKNOWN_TRANSFER: Unknown transfer/run ID: 9849d0e3-4ae5-4e7f-bb29-262e97812440
transfer> exit
Goodbye.
(base) mozany@mozas-MacBook-Air person4-natural-language-success-20260928-011843 %

```

Figure 18: Successful Transfer Under Packet Loss

Despite the introduced packet loss, the receiver successfully completed and verified the file transfer. The reliability mechanisms recovered from the simulated losses, allowing the complete file to be reconstructed and verified at the receiver.

```
(base) mozany@mozas-MacBook-Air person4-natural-language-success-20260928-011843 ~ % find "$RUN_ROOT"/logs/standalone/reconciled -name "summary.json" -print -exec cat {} \;
/Users/mozany/Downloads/udp-file-transfer-integrated-feature-metrics-llm-integration/target/evaluation/person4-natural-language-success-20260928-011843/logs/standalone/reconciled/d07175fd-d5c6-46c1-8980-1b314300aeb9/summary.json
{"schema_version": "1", "metric_definition_version": "1", "evidence_source": "REAL", "reconciliation_status": "VERIFIED", "evidence_completeness": "COMPLETE", "protocol_transfer_id": "d07175fd-d5c6-46c1-8980-1b314300aeb9", "sender_run_id": "9049d0e3-4ae5-4e7f-bb29-262e97812440", "receiver_run_id": "run-6c14faa5-7593-44e5-a354-a71d9ddf2e65", "application_transfer_id": "9049d0e3-4ae5-4e7f-bb29-262e97812440", "experiment_id": null, "metrics": {"experiment_id": null, "file_size_bytes": 262267, "payload_bytes_delivered": 262267, "transfer_time_sec": 0.117496667, "throughput_mbps": 17.856983126168167, "transfer_success": true, "integrity_verified": true, "failure_reason": null, "packets_sent": 257, "packets_received": 257, "packets_dropped": 0, "retransmissions": 0, "acks_received": 257, "packets_acked": 257, "packets_timed_out": 0, "packets_duplicated": 0, "retransmission_ratio": 0.0, "udp_payload_bytes_emitted": 276729, "protocol_overhead_bytes": 14462, "protocol_overhead_ratio": 0.052260514799677665, "rtt_sample_count": 257, "rtt_mean_ms": 2.5661569610894963, "rtt_p95_ms": 6.801125, "chunk_size_bytes": 1024, "window_bytes_requested": 8192, "window_packets": 8, "timeout_ms": 500, "retry_limit": 5, "packet_loss_rate": 0.0, "delay_ms": 0.0, "scenario": "natural-language-success"}, "impairment_seed": 42, "impairment_mechanism": "RECEIVE_DELIVERY_VA", "schema_version": "1", "metric_definition_version": "1", "run_id": null, "application_transfer_id": "9049d0e3-4ae5-4e7f-bb29-262e97812440", "protocol_transfer_id": "d07175fd-d5c6-46c1-8980-1b314300aeb9", "evidence_source": "REAL", "endpoint_attribution": "COMBINED", "file_attribution": null, "capture_timestamp": "2026-09-27T21:24:09.186099Z", "finalization_timestamp": "2026-09-27T21:24:09.186099Z", "integrity_evidence_source": "RECEIVER", "unavailable_reasons": {}, "source_evidence": {"sender_endpoint_record": ".../run-6c14faa5-7593-44e5-a354-a71d9ddf2e65/endpoint-sender.json", "receiver_endpoint_record": ".../run-6c14faa5-7593-44e5-a354-a71d9ddf2e65/endpoint-receiver.json", "sender_event_log": ".../9049d0e3-4ae5-4e7f-bb29-262e97812440/events-sender.json", "receiver_event_log": ".../run-6c14faa5-7593-44e5-a354-a71d9ddf2e65/events-receiver.json", "sender_run_state": ".../9049d0e3-4ae5-4e7f-bb29-262e97812440/run-state.json", "receiver_run_state": ".../run-6c14faa5-7593-44e5-a354-a71d9ddf2e65/run-state.json", "finalized_at_utc": "2026-09-27T21:24:09.186099Z"}}
```

Figure 19: Impaired Transfer Metrics

The completed impaired run recorded the effects of the 2% packet-loss condition. The transfer delivered the complete 262,267-byte file with verified integrity, while recording two simulated packet drops, 16 retransmissions, and two timeout events. The reconciled goodput was approximately 1.80 Mbps.

```
[receiver] Event log: /Users/mozany/Downloads/udp-file-transfer-integrated-feature-metrics-llm-integration/target/evaluation/person4-natural-language-success-20260928-011843/logs/standalone/run-6c14faa5-7593-44e5-a354-a71d9ddf2e65/events-receiver.json
[receiver] Saved /Users/mozany/Downloads/udp-file-transfer-integrated-feature-metrics-llm-integration/target/evaluation/person4-natural-language-success-20260928-011843/storage/incoming/received.bin
[receiver] SUCCESS: Transfer completed and verified
(base) mozany@mozas-MacBook-Air person4-natural-language-success-20260928-011843 % SENDER_DIR=$(find "$RUN_ROOT"/logs/standalone -mindepth 1 -maxdepth 1 -type d -exec test -f '{}' endpoint-sender.json ';' -print)
RECEIVER_DIR=$(find "$RUN_ROOT"/logs/standalone -mindepth 1 -maxdepth 1 -type d -exec test -f '{}' endpoint-receiver.json ';' -print)

echo "SENDER: $SENDER_DIR"
echo "RECEIVER: $RECEIVER_DIR"
SENDER: /Users/mozany/Downloads/udp-file-transfer-integrated-feature-metrics-llm-integration/target/evaluation/person4-natural-language-success-20260928-011843/logs/standalone/9049d0e3-4ae5-4e7f-bb29-262e97812440
RECEIVER: /Users/mozany/Downloads/udp-file-transfer-integrated-feature-metrics-llm-integration/target/evaluation/person4-natural-language-success-20260928-011843/logs/standalone/run-6c14faa5-7593-44e5-a354-a71d9ddf2e65
(base) mozany@mozas-MacBook-Air person4-natural-language-success-20260928-011843 %
transfer_id=9049d0e3-4ae5-4e7f-bb29-262e97812440; run_id=9049d0e3-4ae5-4e7f-bb29-262e97812440; protocol_transfer_id=unavailable
Requested window: 8192 bytes; effective window: 8 packets (8192 bytes); timeout: 500 ms; retry limit: 5 consecutive rounds without progress
transfer> status last
[REAL] COMPLETED; snapshot_at=2026-09-27T21:20:24.662531Z
transfer_id=9049d0e3-4ae5-4e7f-bb29-262e97812440; run_id=9049d0e3-4ae5-4e7f-bb29-262e97812440; protocol_transfer_id=d07175fd-d5c6-46c1-8980-1b314300aeb9
engine_lifecycle=SUCCEEDED; evidence_scope=SENDER_ENDPOINT_LOCAL; evidence_finality=FINAL; provenance=REAL
file_size_bytes=262267
sender_confirmed_acked_payload_bytes=262267
sender_ack_progress_percent=100
live_elapsed_time_sec=0.117496667
sender_ack_based_rate_mbps=17.856983126168167
packets_sent=257
retransmissions=0
acks_received=257
distinct_packets_acked=257
packets_timed_out=0
rtt_sample_count=257
rtt_mean_ms=2.5661569610894963
rtt_p95_ms=6.801125
endpoint_local_udp_payload_bytes_emitted=269542
endpoint_emission_accounting_complete=true
transfer>
```

Figure 20: Impaired Transfer Reconciliation

Sender and receiver evidence from the impaired run was reconciled to confirm the final outcome and combine the measurements from both endpoints. The resulting summary reported the transfer as COMPLETE and VERIFIED and preserved the configured impairment parameters together with the observed reliability and performance metrics.

```
person4-natural-language-success-20260928-011843 — zsh — 80x24
java \
-Dnettransfer.impairment.enabled=true \
-Dnettransfer.impairment.lossPercent=0 \
-Dnettransfer.impairment.delayMs=0 \
-Dnettransfer.impairment.seed=42 \
-Dnettransfer.impairment.scenario=natural-language-success \
-Dnettransfer.logsRoot="$RUN_ROOT/logs" \
-Dnettransfer.receiverInitialTimeoutMs=240000 \
-Dnettransfer.receiverInactivityTimeoutMs=15000 \
-jar "$JAR" receiver 9000 received.bin
[impairment] natural-language-success: receiver DATA loss=0.0%, DATA/ACK delivery delay=0 ms per direction, seed=42; START/FINISH unaffected
[receiver] Binding UDP socket on port 9000 ...
[receiver] Waiting for a transfer...
[receiver] Event log: /Users/mozany/Downloads/udp-file-transfer-integrated-feature-metrics-llm-integration/target/evaluation/person4-natural-language-success-20260928-011843/logs/standalone/run-6c14faa5-7593-44e5-a354-a71d9ddf2e65/events-receiver.jsonl
[receiver] Saved /Users/mozany/Downloads/udp-file-transfer-integrated-feature-metrics-llm-integration/target/evaluation/person4-natural-language-success-20260928-011843/storage/incoming/received.bin
[receiver] SUCCESS: Transfer completed and verified
(base) mozany@mozas-MacBook-Air person4-natural-language-success-20260928-011843
%

person4-natural-language-success-20260928-011843 — java -Dnettran...
Requested window: 8192 bytes; effective window: 8 packets (8192 bytes); timeout: 500 ms; retry limit: 5 consecutive rounds without progress
transfer> status last
[REAL] COMPLETED; snapshot_at=2026-09-27T21:20:24.662531Z
transfer_id=9049d0e3-4ae5-4e7f-bb29-262e97812440; run_id=9049d0e3-4ae5-4e7f-bb29-262e97812440; protocol_transfer_id=d07175fd-d5c6-46c1-8980-1b314300aeb9
engine_lifecycle=SUCCEEDED; evidence_scope=SENDER_ENDPOINT_LOCAL; evidence_finality=FINAL; provenance=REAL
file_size_bytes=262267
sender_confirmed_acked_payload_bytes=262267
sender_ack_progress_percent=100
live_elapsed_time_sec=0.117496667
sender_ack_based_rate_mbps=17.856983126168167
packets_sent=257
retransmissions=0
acks_received=257
distinct_packets_acked=257
packets_timed_out=0
rtt_sample_count=257
rtt_mean_ms=2.5661569610894963
rtt_p95_ms=6.801125
endpoint_local_udp_payload_bytes_emitted=269542
endpoint_emission_accounting_complete=true
transfer>
```

Figure 21: Final Impaired-Run Results

The final results confirmed that the reliable UDP implementation successfully recovered from the controlled packet-loss impairment without corrupting the transferred file. Compared with an unimpaired transfer, the impaired run required retransmissions and timeout-based recovery, providing direct evidence of the protocol's reliability mechanisms under packet loss.